

Unit 1: Foundations and Prerequisites

Prepared by Staples Education

Practice Problems and Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They are graded from raw calculus review, through the number e and Euler's formula, into classifying equations, verifying solutions, and finally building equations by eliminating constants. Complete solutions follow in Part III.

1. Differentiate $y = e^{3x}$ using the chain rule.
 2. Evaluate the indefinite integral $\int e^{-2x} dx$.
 3. Differentiate **(a)** $(2x + 1)^3$ and **(b)** $\sin(x^2)$.
 4. Compute $\frac{d}{dt}e^{kt}$, then state the general solution of $\frac{dy}{dt} = y$.
 5. Use Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ to evaluate **(a)** $e^{i\pi}$ and **(b)** $e^{i\pi/2}$, and write down Euler's identity.
 6. For each equation, state the order and the degree, and say whether it is linear: **(a)** $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} = 0$, **(b)** $\left(\frac{dy}{dx}\right)^3 + y = x$, **(c)** $\frac{dy}{dx} + y^2 = 0$.
 7. Verify that $y = e^{2x}$ is a solution of $\frac{d^2y}{dx^2} - 4y = 0$.
 8. Decide whether $y = x^2$ satisfies $x\frac{dy}{dx} = 2y$, and whether $y = \cos x$ satisfies $\frac{d^2y}{dx^2} + y = 0$.
 9. Form a differential equation by eliminating C from **(a)** $y = Cx^2$ and **(b)** $y = Ce^x$.
 10. **(Capstone)** Eliminate both constants from $y = A \cos x + B \sin x$, and separately eliminate r from the circle family $x^2 + y^2 = r^2$.
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Part III — Complete Solutions

Solution 1. The exponential reproduces itself, and the chain rule multiplies by the derivative of the inside, 3.

$$\frac{dy}{dx} = e^{3x} \cdot 3 = 3e^{3x}.$$

Solution 2. Integrating an exponential divides by the constant in the exponent and restores the arbitrary constant.

$$\int e^{-2x} dx = \frac{1}{-2} e^{-2x} + C = -\frac{1}{2} e^{-2x} + C.$$

Solution 3. Apply the power-and-chain rule, then the chain rule on the sine.

$$\frac{d}{dx}(2x+1)^3 = 3(2x+1)^2 \cdot 2 = 6(2x+1)^2$$

$$\frac{d}{dx} \sin(x^2) = \cos(x^2) \cdot 2x = 2x \cos(x^2).$$

Solution 4. The exponential returns itself times the constant from the exponent.

$$\frac{d}{dt} e^{kt} = k e^{kt}.$$

The equation $\frac{dy}{dt} = y$ asks for a function equal to its own derivative, so

$$y = C e^t.$$

Solution 5. Substitute the angles into Euler's formula.

$$e^{i\pi} = \cos \pi + i \sin \pi = -1 + 0 = -1$$

$$e^{i\pi/2} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = 0 + i = i.$$

Rearranging the first gives Euler's identity, $e^{i\pi} + 1 = 0$.

Solution 6. The order is the highest derivative; the degree is its power; linearity forbids powers above one on y and its derivatives.

(a) order 2, degree 1, linear

(b) order 1, degree 3, nonlinear

(c) order 1, degree 1, nonlinear (the y^2 term).

Solution 7. Differentiate twice and substitute.

$$\frac{dy}{dx} = 2e^{2x}, \quad \frac{d^2y}{dx^2} = 4e^{2x}$$

$$\frac{d^2y}{dx^2} - 4y = 4e^{2x} - 4e^{2x} = 0 \quad \checkmark$$

so $y = e^{2x}$ is a solution.

Solution 8. Test each candidate by substitution.

$$y = x^2 : \quad y' = 2x, \quad x y' = x \cdot 2x = 2x^2 = 2y \quad \checkmark$$

$$y = \cos x : \quad y'' = -\cos x, \quad y'' + y = -\cos x + \cos x = 0 \quad \checkmark$$

Both candidates check out.

Solution 9. Differentiate, then replace the constant from the original relation.

$$y = Cx^2 : \quad y' = 2Cx, \quad C = \frac{y}{x^2} \implies y' = 2 \cdot \frac{y}{x^2} \cdot x = \frac{2y}{x}$$

$$\text{so } x \frac{dy}{dx} = 2y.$$

$$y = Ce^x : \quad y' = Ce^x = y \implies \frac{dy}{dx} = y.$$

Solution 10. Each family needs differentiation to clear its constants. For $y = A \cos x + B \sin x$ (two constants), differentiate twice:

$$y'' = -A \cos x - B \sin x = -y \implies \frac{d^2y}{dx^2} + y = 0.$$

For the circle $x^2 + y^2 = r^2$ (one constant), differentiate implicitly:

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

Unit 1 Key Takeaway

These problems exercise the two directions of the subject. Going forward, you **differentiate and integrate** fluently and **verify a solution** by substituting it into the equation; going backward, you **eliminate arbitrary constants** to build the equation a family satisfies. Throughout, the **order, degree, and linearity** of an equation are the first labels you read off.